Automated Building Validator - Week 5

Green Metrics Technology (GMT)

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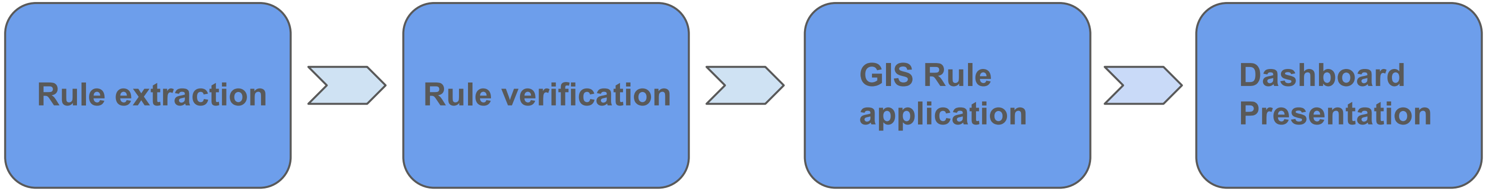
Current Project Outline

1. Rule Extraction Pipeline that extract and normalize all the rules from PDF bylaw files
2. Verification Layer to verify all the extracted rules and send the verified ones to GIS layer
3. The GIS layer take in the rules and apply them to city lots
4. The final product will be shown in a dashboard with the GIS map.



Rule Extraction Layer: Zihao

Update since last week:

The rule extraction is working properly on the Burnaby case, the challenges are:

1. Too high API cost
2. Not generalized enough for other cities

Our partner mentioned the ‘tree case’, bylaws stating if a tree can be cut down from the lot for building the home. No such case for Burnaby.

Rule Extraction Layer: Zihao

Main upgrades:

1. Reduced the API token cost by 50%
2. Generalize the model so that it can look for these ‘tree cases’
3. Experimenting on new table extraction algorithm since it’s the most expensive part (still in development)

	Dwelling Type			
	Rowhouse	Small-Scale Multi-Unit		
Permitted Dwelling Units (including secondary suites)	1 to 3 Units	1 to 2 Units	3 to 4 Units	5 to 6 Units Frequent Transit Network Area Only
Minimum Lot Area	-	-	4 Units Only: 281 m^2	281 m^2
Maximum Lot Area [1]	280 m^2	-	-	-
Maximum Lot Coverage				
All Buildings	55%	Lots ≤ 567 m^2: 40% Lots > 567 m^2: 30%	40%	45%
Impervious Surfaces	70%	60%	60%	70%
Maximum Height				
Front Principal Buildings				
Height	sloping roof: 10 m | flat roof: 9.5 m			
Storeys (basement inclusive)	3 storeys	2.5 storeys	3 storeys	
Rear Principal Buildings				
Height	sloping roof: 7.5 m | flat roof: 7.0 m			
Storeys (basement inclusive)	2 storeys			
Accessory Buildings	4.0 m | 1 storey			
Minimum Lot Line Setbacks for All Buildings [2,3]				
Street Yard	3.0 m	front: 4.0 m | flanking: 3.0 m		
Lane Yard	1.5 m			
Interior Rear Yard	3.0 m, except 1.5 m for accessory buildings			
Interior Side Yard	0 m, except 1.2 m for end unit lots	1.2 m	1.2 m	1.2 m
Minimum Separation of Buildings on the Same Lot [4,5]				
Between Front Principals	-	2.4 m	2.4 m	2.4 m
Between Rear Principals	-	2.4 m	2.4 m	2.4 m
Between Front & Rear Principals	6.0 m			
Between All Other Buildings	2.4 m			

Convert the table to csv, split merged cell, then scan through value cell one-by-one, take consideration of table color this time, since it shows which belongs to which.

Rule Verification: Yusen

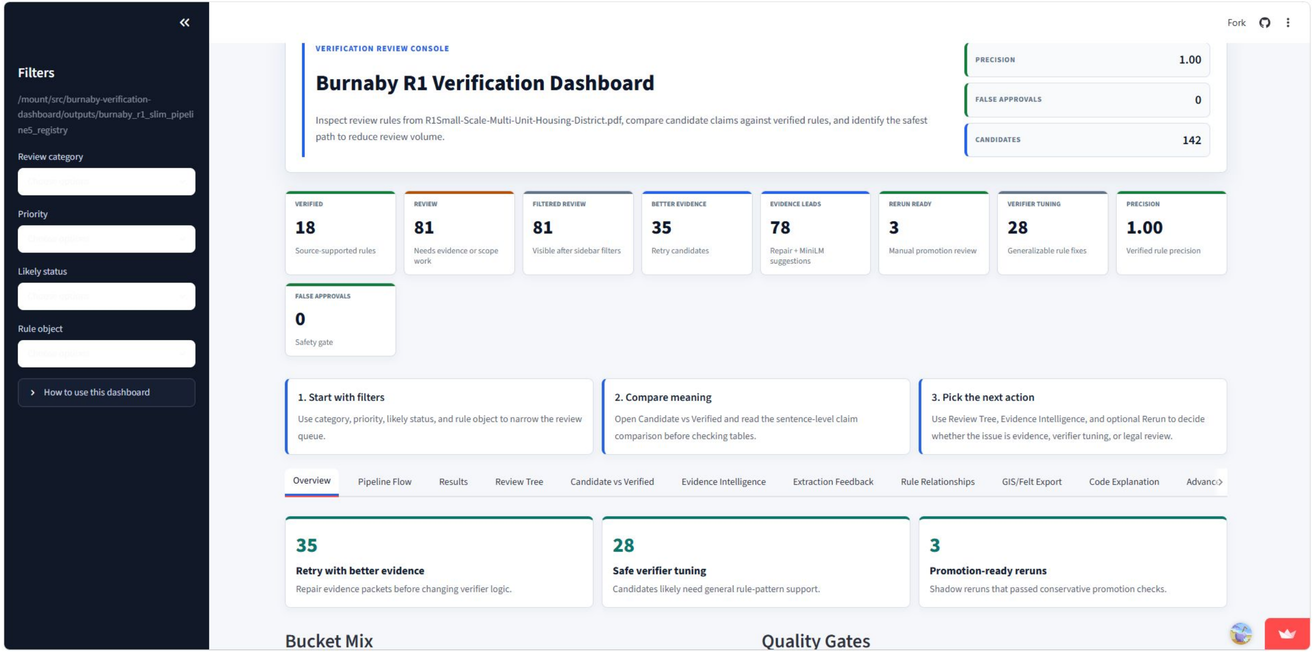
Update since last week:

Last week, the extracted rules was sending to many true rules to rejected and review. The problem is being fixed gradually:

1. Instead of rejecting correct but not important rules, the rules are now assigned to ‘not used’ section
2. Added candidate-evidence comparison and evidence repair support for reviewed rules.

Rule Verification: Yusen

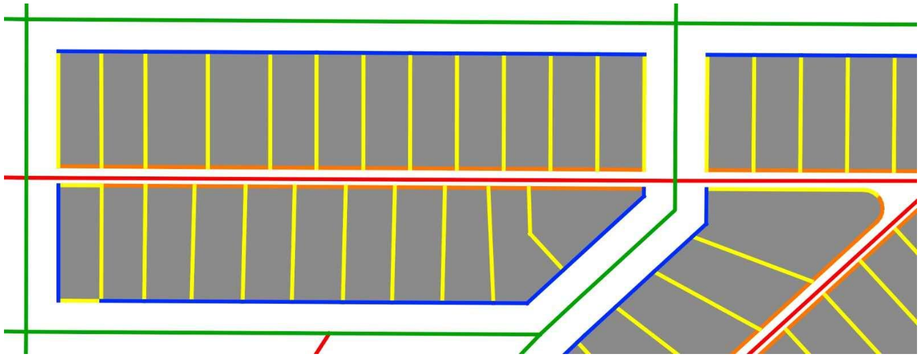
To make the verification procedure clearer and result easier to track, a dashboard is created to keep track of the quality and number of the rules



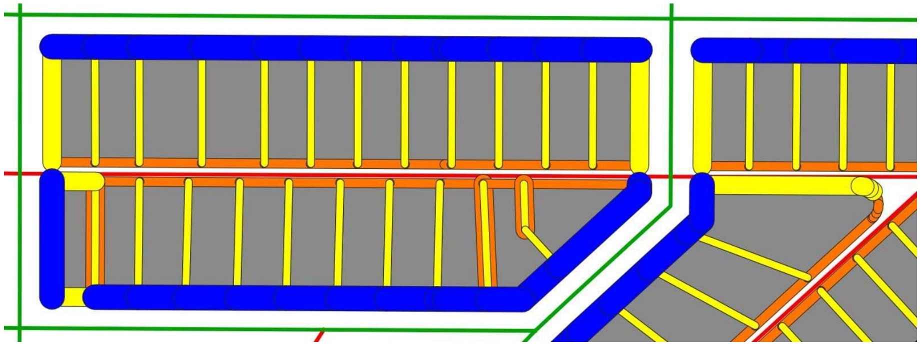
GIS Rule Application: Jamie

The lot line labelling algorithm is applied to 1000 R1 lots in Burnaby:

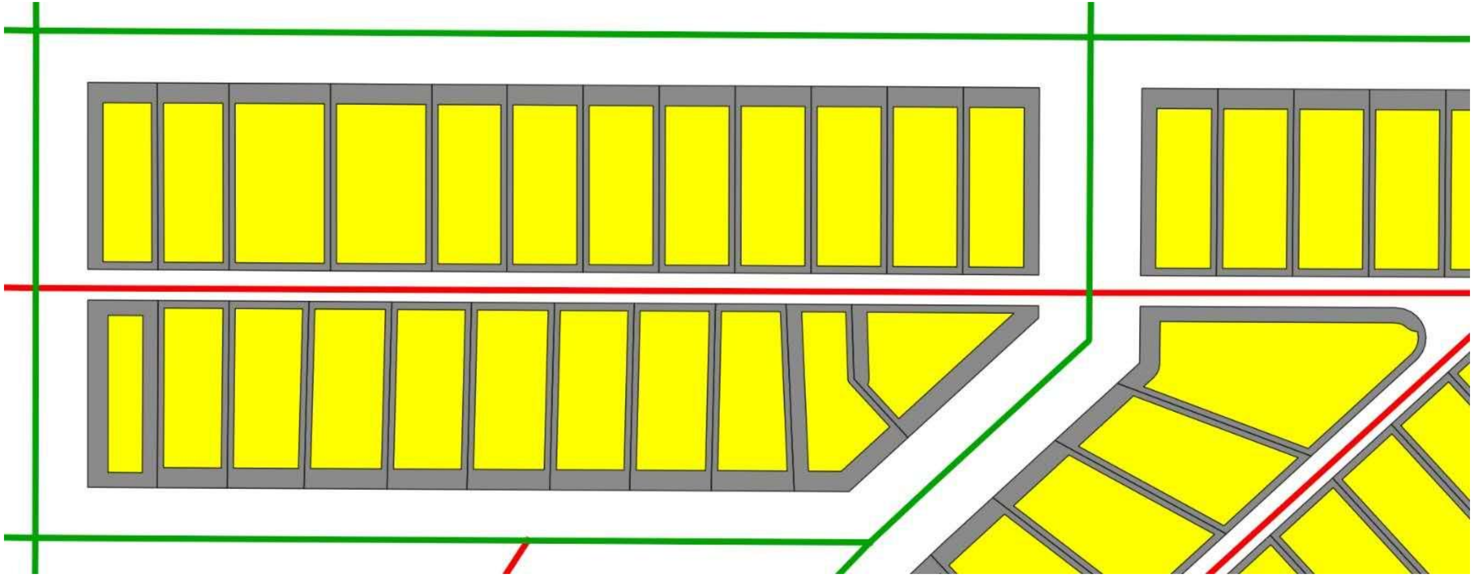
Front: Blue
Side: Yellow
Back: Orange



Setback rules are then applied to each based on label

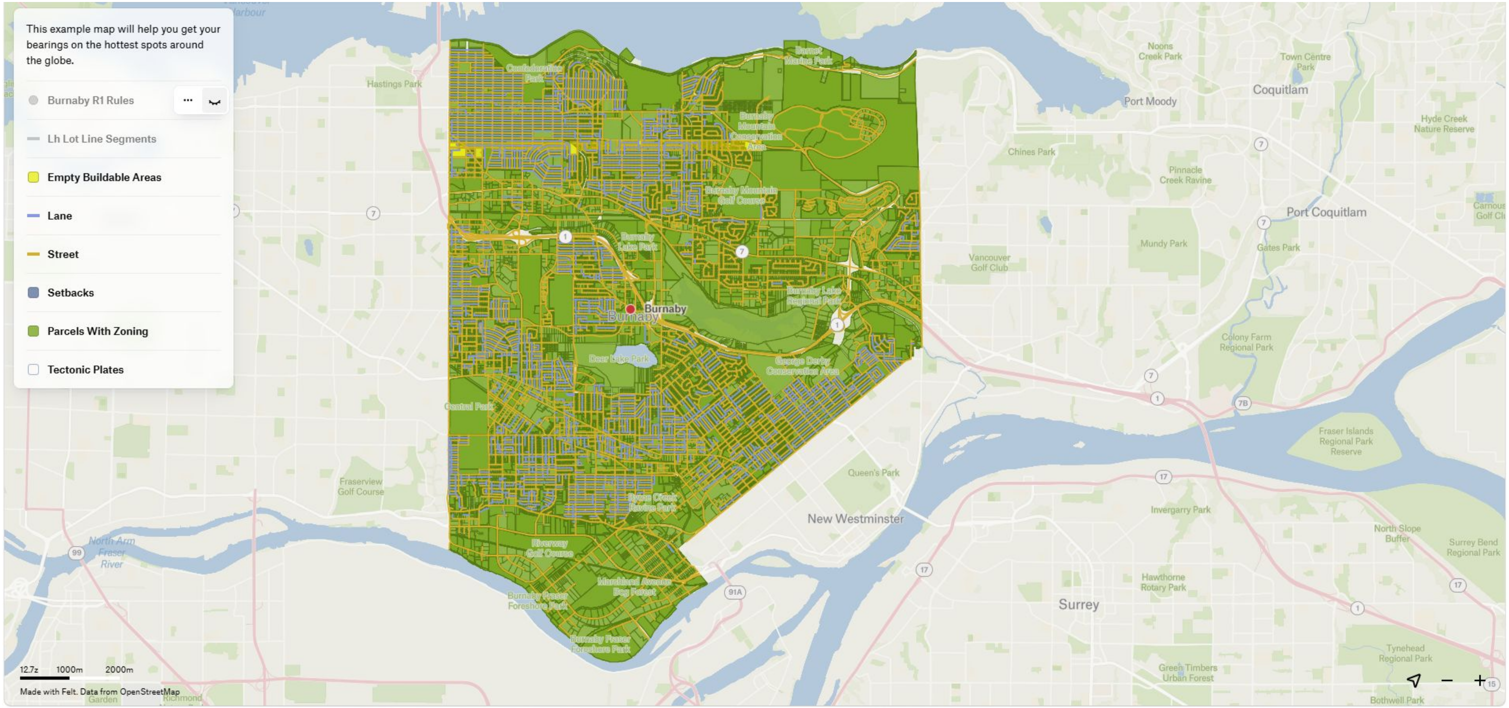


GIS Rule Application: Jamie



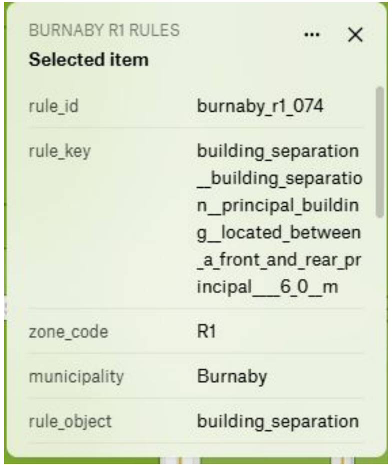
Final buildable area is shown as yellow
* Currently, existing structures are not taken into account. That will be added this week.

Dashboard: Sasivimol



Dashboard Overview

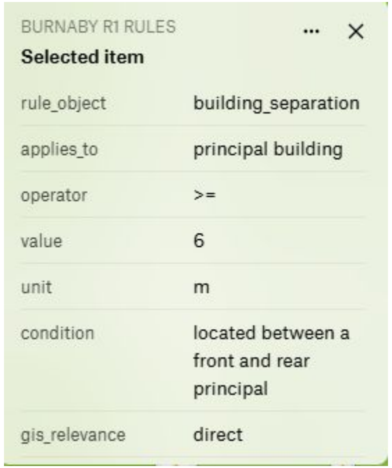
Dashboard: Sasivimol



BURNABY R1 RULES

Selected item

rule_id	burnaby_r1_074
rule_key	building_separation__building_separation_principal_building_located_between_a_front_and_rear_principal___6_0_m
zone_code	R1
municipality	Burnaby
rule_object	building_separation

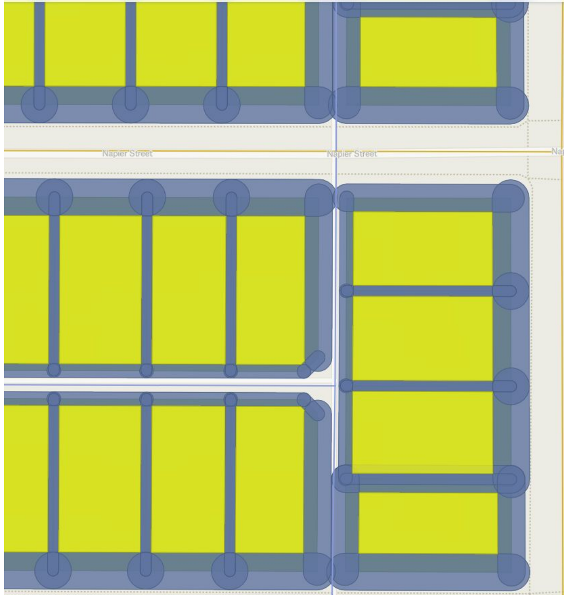


BURNABY R1 RULES

Selected item

rule_object	building_separation
applies_to	principal building
operator	>=
value	6
unit	m
condition	located between a front and rear principal
gis_relevance	direct

Rules are now imported and shown in the dashboard



Basic setback rules are now applied to lots (Blue areas) and the buildable areas (yellow areas) are shown in the map

Next Steps

1. Rule extraction pipeline cost and efficiency optimization
2. Rule verification layer accuracy improvement
3. Adding existing structures to the GIS system
4. Dashboard layout improvement and further development

Work Summary

Team Member	Primary Tasks	Hours	Team Member	Primary Tasks	Hours
Jamie	1. Developing a new lot buildable area algorithm for parcels of different shapes 2. Implemented the empty-lot buildable area algorithm in QGIS for 1000 Burnaby lots	**40**	**Thomas**	1. Improved verification workflow with clearer review routing and evidence repair support. 2. Updated Streamlit dashboard and clarified verified-only GIS/Felt handoff.	**40**
Bee	1. Created an interactive Felt map for Burnaby with parcels, zoning overlays, buildable areas, and reference layers 2. Integrated the rules lookup table into Felt for dynamic rule display when users click parcels	**33**	**Zihao**	1. Continued optimizing the rule extraction pipeline and summarized current extraction results 2. Ran new table extraction experiments and researched PIBC’s existing rule extraction system	**40**